

Midpoint preservation under lens containment

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

theorem

Restricts a set isometry and its inverse to the two contained lenses.

Statement

Let E, F be real normed spaces, $S \subseteq E$, $T \subseteq F$, and $f : S \rightarrow T$ a bijective isometry. Choose $x, y \in S$ and $r \in \mathbb{R}$. Write $\Lambda_r(x, y) = \overline{B}(x, r) \cap \overline{B}(y, r)$. If

$$\frac{\|x - y\|}{2} \leq r, \quad \Lambda_r(x, y) \subseteq S, \quad \Lambda_r(f(x), f(y)) \subseteq T,$$

then $f((x + y)/2) = (f(x) + f(y))/2$.

Assumptions

The half-distance bound and both displayed lens inclusions are assumed for the same points, radius and isometry. There is no openness or convexity requirement on S, T .

Conclusion

The midpoint belongs to S and its image is the midpoint of the two images. The target half-distance bound is a consequence of distance preservation.

Proof route

Use both inclusions to restrict the forward and inverse maps; then apply the lens midpoint theorem to that restriction.

▼ Proof steps

1. If $z \in \Lambda_r(x, y)$, then $z \in S$ and

$$\|f(z) - f(x)\| = \|z - x\| \leq r, \quad \|f(z) - f(y)\| = \|z - y\| \leq r.$$

Hence $f(z)$ lies in the target lens. Conversely, for w in that lens, the target inclusion allows $f^{-1}(w)$ to be formed, and the same two distance identities put it in the source lens. This is precisely the bijective isometry in Restriction to corresponding lenses.

2. Distance preservation gives $\|f(x) - f(y)\|/2 = \|x - y\|/2 \leq r$. The source half-distance bound and Midpoint membership also give $(x + y)/2 \in \Lambda_r(x, y) \subseteq S$.

3. Apply Midpoint preservation between lenses to the restricted isometry, source foci x, y , target foci $f(x), f(y)$ and radius r . Both half-distance hypotheses were checked above. Since this restriction evaluates to the original f , its output is $f((x + y)/2) = (f(x) + f(y))/2$.

Main citations

- Exact theorem and proof
- Restriction to corresponding lenses
- Midpoint preservation between lenses
- Lens definition
- Midpoint membership

Lean source signature (exact)

```
theorem map_midpoint_of_symmetricLens_subset
  {s : Set E} {t : Set F} (f : s  $\simeq_i$  t) (x y : s) (r :  $\mathbb{R}$ )
  (hhalf :  $2^{-1} * \text{dist } (x : E) (y : E) \leq r$ )
  (hs : symmetricLens (x : E) (y : E) r  $\subseteq$  s)
  (ht : symmetricLens ((f x : t) : F) ((f y : t) : F) r  $\subseteq$  t) :
  ((f (midpoint  $\mathbb{R}$  (x : E) (y : E),
    hs (midpoint_mem_symmetricLens hhalf)) : t) : F) =
  midpoint  $\mathbb{R}$  ((f x : t) : F) ((f y : t) : F)
```

In the source	Mathematical meaning
$(f : s \simeq_i t) (x y : s) (r : \mathbb{R})$	The bijective isometry $f : S \rightarrow T$, two points x, y already in S , and a real radius r . The ambient E, F are real normed spaces.
$(hhalf : 2^{-1} * \text{dist } (x : E) (y : E) \leq r)$	The source half-distance condition $\ x - y\ /2 \leq r$; the casts use the same points in E .
$\text{symmetricLens } (x : E) (y : E) r \subseteq s$	Every point satisfying both source closed-ball inequalities belongs to S .
$\text{symmetricLens } ((f x : t) : F) ((f y : t) : F) r \subseteq t$	The entire lens with foci $f(x), f(y)$ and the same radius belongs to T .
$hs (\text{midpoint_mem_symmetricLens } hhalf)$	The derived proof that $(x + y)/2 \in S$: first use the half-distance bound to enter the lens, then its inclusion in S .
$((f (\text{midpoint } \mathbb{R} (x : E) (y : E), hs (\text{midpoint_mem_symmetricLens } hhalf)) : t) : F)$	The well-defined point $f((x + y)/2)$ regarded in F .
$\text{midpoint } \mathbb{R} ((f x : t) : F) ((f y : t) : F)$	The conclusion equates that point with $(f(x) + f(y))/2$.

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Accepted content SHA-256: `a22ad0229a763a9c4fbb846f8874c408c690b05e0fe21eb51c664fd53bc9e568`

Accepted source guide SHA-256: `745bce0b99b52d2d2f0a7c54c4d5323a3cbafd4ac67bd9500839ed3d4ff6da07`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`